A gamified introduction to Python Programming

Lecture 12
Memory management in lists and Numpy

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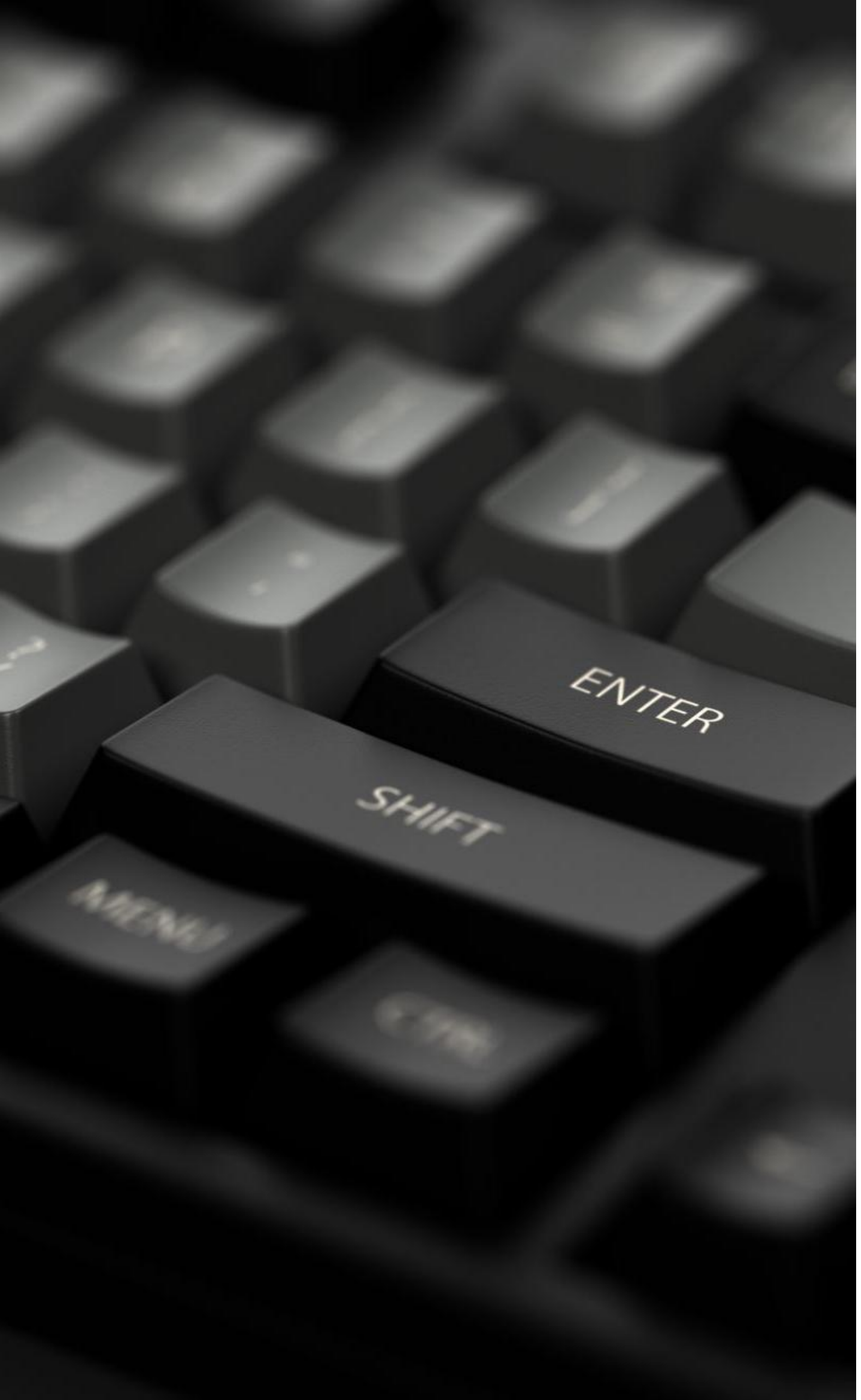

Outline (Chapter 12)

- A quick word on memory management and lists: **aliasing**, **shallow** and **deep copies.**
- What is a **library**?
- What is the **Numpy library** and its basic concepts (e.g. arrays, math functions, etc.)?

Memory of a computer

The memory of a computer consists of several “boxes”, which contain values for variables. Each “box” is identified by an **address/ID.**

When a variable is created:

- A “box” is assigned for the variable and its value is stored in the “box”.
- The variable name simply refers to the address/ID of the “box”.

Identifier **Memory**

Identifier	Address	Value
x1 →	140725454247872	10
	…	…

```python
x1 = 10
print(x1)
print(id(x1))
```

```
10
140725454247872
```

Recall: the id() function

Definition (Aliasing):

Python saves memory space by having two variables **point to the same memory ID/location**.
This is called **aliasing**.

- Two variables with different names but identical values might end up being stored in the same address, to save memory space.
- This can be confirmed by checking the **id()** of both variables.

```python
x1 = 10
print(x1)
print(id(x1))
```

```
10
140725454247872
```

```python
x2 = 17
print(x2)
print(id(x2))
```

```
17
140725454248096
```

```python
x3 = x1
print(x3)
print(id(x3))
print(id(x1))
```

```
10
140725454247872
140725454247872
```

Recall: the id() function

Definition (Aliasing):

Python saves memory space by having two variables **point to the same memory ID/location**.
This is called **aliasing**.

- Two variables with different names but identical values might end up being stored in the same address, to save memory space.
- This can be confirmed by checking the **id()** of both variables.

Identifier | **Memory**

x1, x3 → 140725454247872

Address	Value
140725454247872	10
…	…

```python
x3 = x1
print(x3)
print(id(x3))
print(id(x1))
```

```
10
140725454247872
140725454247872
```

Memory management in lists

A **list** is a collection of variables.

- If variables are assembled in a list, we reuse the existing memory locations (same id).

```
list1 = [x1, x2, x3]
print(list1)
print(id(list1))
```

```
[12, 17, 10]
1769354632448
```

```
print(id(list1))
print("-")
print(id(list1[0]))
print(id(x1))
print("-")
print(id(list1[1]))
print(id(x2))
print("-")
print(id(list1[2]))
print(id(x3))
```

```
1769354632448
-
140725454247936
140725454247936
-
140725454248096
140725454248096
-
140725454247872
140725454247872
```

Memory management in lists

A **list** is a collection of variables.

- The variables in a list are **chained** together.
- (In practice, it is actually more complex than this, but for now, at our current beginner level, this will do!)

Identifier **Memory**

Identifier	Address	Value
list1	1769354632448	List value
	…	…

list1[0]

Address	140725454247936
Value	12
Next Address	140725454247872

list1[1]

Address	140725454247872
Value	17
Next Address	140725454248096

list1[2]

Address	140725454248096
Value	10
Next Address	End of list

Memory management in lists

When playing with simple (non-compound) types of variables, such as int, float and strings, Python is smart enough to address aliasing.

- If x1 is changed, Python will adjust so that the list remains unaffected.
- It simply reallocates x1 to another location in memory, so the list remains unaffected.

```python
print(x1)
print(id(x1))
x1 = "Hello"
print(id(x1))
print(list1)
```

```
12
140726382041088
2120755150000
[12, 17, 10]
```

Aliasing in lists: problem

A **list** is a collection of variables.

- The variables in a list are **chained** together.
- **Aliasing:** We can assign a list to another variable name and reuse the existing memory spaces, not creating new ones.

```
list2 = list1
print(list2)
print(id(list1))
print(id(list2))
```

```
[12, 17, 10]
1983742564288
1983742564288
```

Aliasing in lists: problem

A **list** is a collection of variables.

- The variables in a list are **chained** together.
- **Aliasing:** We can assign a list to another variable name and reuse the existing memory spaces, not creating new ones.

Identifier | **Memory**

Identifier	Address	Value
list1, list2	1769354632448	List value
	…	…

list1[0] list2[0]

Address	140725454247936
Value	12
Next Address	140725454247872

list1[1] list2[1]

Address	140725454247872
Value	17
Next Address	140725454248096

list1[2] list2[2]

Address	140725454248096
Value	10
Next Address	End of list

Aliasing in lists: problem

Problem: changing list1[0] changes list1 values, but also changes list2!

- When aliasing is used on list variables, Python is not smart enough to address the aliasing to leave the second list unaffected!

```python
print(id(list1[0]))
list1[0] = "SUTD"
print(list1)
print(id(list1[0]))
```

```
140726382041088
['SUTD', 17, 10]
2120755353584
```

```python
print(list2)
print(id(list2[0]))
```

```
['SUTD', 17, 10]
2120755353584
```

Aliasing in lists: problem

Problem: changing list1[0] changes list1 values, but also changes list2!

- When aliasing is used on list variables, Python is not smart enough to address the aliasing to leave the second list unaffected!

Identifier	Memory: Address	Memory: Value
list1, list2 →	1769354632448	List value
	…	…

list1[0] list2[0]

Address	140725454247936
Value	12 → "SUTD"
Next Address	140725454247872

list1[1] list2[1]

Address	140725454247872
Value	17
Next Address	140725454248096

list1[2] list2[2]

Address	140725454248096
Value	10
Next Address	End of list

Shallow copy of a list

Definition (shallow copying):

The operation list2 = list1[:] makes list2 a **shallow copy** of list1.
By doing so, list2 will be saved to its own location of memory.

- Having done a shallow copy, changing a value in list1, with list1[index] = …, no longer affects the values list2. They have different memory IDs now!
- **Note:** you can also use the copy() method.

```python
list1 = [12, 17, 10]
list2 = list1[:]
print(list1)
print(list2)
print(id(list1))
print(id(list2))
```

```
[12, 17, 10]
[12, 17, 10]
2120755431296
2120755345920
```

```python
list1[0] = "SUTD"
print(list1)
print(list2)
```

```
['SUTD', 17, 10]
[12, 17, 10]
```

Shallow copy: problem

Note: if an element of a list is a list (case of lists of lists), then the shallow copy will not copy the sublists to different locations of memory.

- **Problem:** changing the element of a sublist then affects both lists, even though these lists are shallow copies of each other…
- The problem repeats!

```python
list1 = [[8, 9, 11], 7, 4]
list2 = list1[:]
print(list1)
print(list2)
print(id(list1))
print(id(list2))
print(id(list1[0][1]))
print(id(list2[0][1]))
```

```
[[8, 9, 11], 7, 4]
[[8, 9, 11], 7, 4]
2120755333248
2120755332928
140726382040992
140726382040992
```

```python
list1[0][1] = "Damn it!"
print(list1)
print(list2)
```

```
[[8, 'Damn it!', 11], 7, 4]
[[8, 'Damn it!', 11], 7, 4]
```

Deep copy

Solution: make a **deep copy**, using the **deepcopy** function from the Python built-in **copy** library.

- A deep copy forces Python to make sure **all elements and sub-elements** are assigned to different locations in memory.
- **Stronger and safer than a shallow copy!**
- **If in doubt, just deep copy!**

```
from copy import deepcopy
```

```
list1 = [[8, 9, 11], 7, 4]
list2 = deepcopy(list1)
print(list1)
print(list2)
print(id(list1[0]))
print(id(list2[0]))
```

```
[[8, 9, 11], 7, 4]
[[8, 9, 11], 7, 4]
2120754851072
2120754850304
```

```
list1[0][1] = "Deep copy works?"
print(list1)
print(list2)
print(id(list1[0][1]))
print(id(list2[0][1]))
```

```
[[8, 'Deep copy works?', 11], 7, 4]
[[8, 9, 11], 7, 4]
2120755409424
140726382040992
```

Matt’s Great advice

Matt’s Great Advice: Keep the aliasing, shallow and deep copies concepts in mind for now.

If you find that modifying a list object ends up unexpectedly changing another, then you might have an aliasing or shallow copy problem.

When in doubt, make a deep copy.

For now, do not worry about understanding all these memory concepts, these will be covered in another advanced course!

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What needs to be changed to make the printed result [[0, 5, 0], [0, 0, 0], [0, 0, 0]]?

ⓘ Start presenting to display the poll results on this slide.

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What will be displayed?

ⓘ Start presenting to display the poll results on this slide.

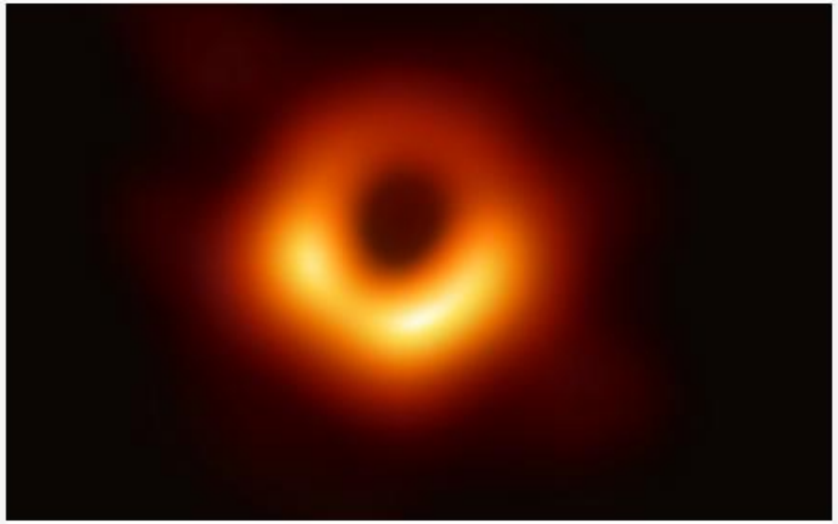
About the numpy library

NumPy

NumPy (or numpy or Numpy) is one of the most common **libraries** (if not the most popular one) in Python.

- It is used for many applications: computing, modelling, data science, astrophysics, etc.
- **Linear algebra** (one of the main concepts of math with many applications in computing)

FIRST IMAGE OF A BLACK HOLE



How NumPy, together with libraries like SciPy and Matplotlib that depend on NumPy, enabled the Event Horizon Telescope to produce the first ever image of a black hole

A new type of objects: Numpy Arrays

Numpy arrays are objects from the **Numpy** library.

- Typically used to describe **matrices** and **vectors,**
- Or **tables** of data.

They look very similar to **lists of lists**, which we have used earlier for many applications.

The Numpy library, however, comes with many additional functions and methods.

```python
import numpy as np
```

```python
array1 = np.array([0, 2, 1, 4])
print(array1)
```

```
[0 2 1 4]
```

```python
print(array1)
print(type(array1))
array1_as_list = list(array1)
print(array1_as_list)
print(type(array1_as_list))
```

```
[0 2 1 4]
<class 'numpy.ndarray'>
[0, 2, 1, 4]
<class 'list'>
```

Length, size, shape

As lists, Numpy arrays have a length, which can be checked with **len()**.

- The also have a **shape** and a **size attribute**, which give additional information, in the case of arrays with more than 1D.
- **Attribute: "sub-variable"** of an object; can be drawn from an object using the **.** operator (Has to do with object-oriented prog., covered later).

```
array1 = np.array([0, 2, 1, 4])
print(len(array1))
print(array1.shape)
print(array1.size)
```

```
4
(4,)
4
```

```
two_d_array = np.array([[1,2],[3,4]])
print(two_d_array)
print(len(two_d_array))
print(two_d_array.shape)
print(two_d_array.size)
```

```
[[1 2]
 [3 4]]
2
(2, 2)
4
```

Indexing an array

Just like lists, the Numpy arrays are indexed and their element can be accessed with [].

- You can equivalently use the [i,j] and [i][j] notations on arrays.
- Replacing an index with a colon symbol :, means “take all”.
- For instance, [:, j] means **all elements in column j**, whereas [i, :] means **all elements in row i**.

```
array1 = np.array([0, 2, 1, 4])
print(array1)
print(array1[0])
print(array1[1])
```

```
[0 2 1 4]
0
2
```

```
two_d_array = np.array([[1,2],[3,4]])
print(two_d_array)
print(two_d_array[0])
print(two_d_array[0][1])
```

```
[[1 2]
 [3 4]]
[1 2]
2
```

```
print(two_d_array[0,1])
print(two_d_array[:,1])
print(two_d_array[0,:])
```

```
2
[2 4]
[1 2]
```

Traversing an array with for

As with **lists**, we can traverse a Numpy array, in an element-wise manner, using a **for** loop.

```
array1 = np.array([0, 2, 1, 4])
print(array1)
for element in array1:
    print(element)
```

```
[0 2 1 4]
0
2
1
4
```

```
my_list = [1, 4, 9, 14, 15]
print(my_list)
```

```
[1, 4, 9, 14, 15]
```

```
# Element-wise
for element in my_list:
    print("--")
    print(element)
```

```
--
1
--
4
--
9
--
14
--
15
```

The + operator on arrays

- **The + operator on lists**: On lists, the + operator will **concatenate** both lists into a new one.
- **The + operator on Numpy arrays – (vector sum):** On Numpy arrays, however, the + operator will **sum** the elements of both Numpy arrays.
- **Broadcasting:** If summed with a number instead, the elements in the Numpy array will each be incremented by the given value.

```
a_list = [0, 1, 2]
another_list = [1, 4, 7]
list_sum = a_list + another_list
print(list_sum)
```

```
[0, 1, 2, 1, 4, 7]
```

```
array1 = np.array([0, 2, 1, 4])
array2 = np.array([1, 2, 3, 5])
print(array1)
print(array2)
sum_array = array1 + array2
print(sum_array)
```

```
[0 2 1 4]
[1 2 3 5]
[1 4 4 9]
```

```
array1 = np.array([0, 2, 1, 4])
number = 7
print(array1)
sum_array = array1 + number
print(sum_array)
```

```
[0 2 1 4]
[ 7  9  8 11]
```

Aliasing, Shallow and Deep copies in arrays

- As with lists, Numpy arrays are subject to the same issues about **aliasing, shallow and deep copies**.

- If needed, use deep copies of the arrays.

```
two_d_array1 = np.array([[1,2],[3,4]])
two_d_array2 = two_d_array1
print(two_d_array1)
print(two_d_array2)
print(id(two_d_array1))
print(id(two_d_array2))
```

```
[[1 2]
 [3 4]]
[[1 2]
 [3 4]]
1750029422960
1750029422960
```

```
two_d_array1[0][0] = 17
print(two_d_array1)
print(two_d_array2)
```

```
[[17  2]
 [ 3  4]]
[[17  2]
 [ 3  4]]
```

Activity 1 - Exam adjustments

- Let us assume, that I have **grades** from my students listed in some **Numpy array** variable, as shown below.

```
columns_labels_list = ["MidTerm Score", "FinalExam Score", "Average Score"]
students_list = ["Chris", "Oka", "Norman", "Natalie", "Tony"]
grades_table = np.array([[60, 80, 70], \ # Chris scored 60% on MidTerm, 80% on Finals, Average is 70%
                         [50, 80, 65], \
                         [40, 70, 55], \
                         [60, 70, 65], \
                         [60, 90, 75]])
print(grades_table)
```

```
[[60 80 70]
 [50 80 65]
 [40 70 55]
 [60 70 65]
 [60 90 75]]
```

Activity 1 - Exam adjustments

- Let us assume, that I have **grades** from my students listed in some **Numpy array** variable.
- The first line contains the column labels (student name, some scores) and the other lines will consist of entries regarding some of the students.
- Let us assume that, as a professor, I have decided to be lenient towards my students.
- I realized that the midterm was a bit too difficult compared to last year, and they failed more than expected… To compensate for that, I would like to increase the scores of all students on the midterm by 50%.
- Will use a function to do so!

Activity 1 - Exam adjustments

Write a function **grade_adjustment()**,

- which **receives** a grades table, **grades_table**,
- **increases** the **scores** of all students on **midterm** by **50%**,
- **re-calculates** the **average score**, with the **new adjusted midterm score**,
- and then returns a <u>**new**</u> **updated grades table** as its sole output.

- **Important note:** The **maximal score** for the midterm exam is **capped** to **100**. This means that a student which scores 80 points on the midterm, will not obtain 120 points after the adjustment, but only 100.

Concatenation on arrays

- Since the **+** operator cannot be used for **concatenation**, Numpy comes with a **concatenate()** function.

```
print(array1)
print(array2)
conc_array = np.concatenate([array1, array2])
print(conc_array)
```

```
[0 2 1 4]
[1 2 3 5]
[0 2 1 4 1 2 3 5]
```

The * operator on arrays

- The * operator behaves as the + operator on Numpy arrays.
- It consists of an **element-wise multiplication** of the elements in arrays.
- **Broadcasting:** if a Numpy array is multiplied by a number, the number will multiply each element in the array.

```
array1 = np.array([0, 2, 1, 4])
array2 = np.array([1, 2, 3, 5])
print(array1)
print(array2)
mult_arrays = array1*array2
print(mult_arrays)
```

```
[0 2 1 4]
[1 2 3 5]
[ 0  4  3 20]
```

```
n = 4
mult_array_int = array1*n
print(mult_array_int)
```

```
[ 0  8  4 16]
```

Additional functions

Additional Numpy functions

- **Min, max:** returns the minimal, resp. maximal, values in array.
- **Argmin, argmax:** returns the index where the minimal, resp. maximal, values are.
- **Mean, median:** returns the mean, resp. median, value for a given array.
- **Sum:** sums all the elements in the array together

```
array1 = np.array([0, 2, 1, 4, 7])
print(array1)
min_val = np.min(array1)
print(min_val)
argmin_val = np.argmin(array1)
print(argmin_val)
max_val = np.max(array1)
print(max_val)
argmax_val = np.argmax(array1)
print(argmax_val)
mean_val = np.mean(array1)
print(mean_val)
median_val = np.median(array1)
print(median_val)
summed_val = np.sum(array1)
print(summed_val)
```

```
[0 2 1 4 7]
0
0
7
4
2.8
2.0
14
```

Mathematical functions and constants

Numpy also contains

- Many **mathematical functions** (cosine, sine, logarithm, exponential, etc.)
- And many **mathematical constants** (pi, etc.)

```
print(np.cos(0))
print(np.sin(0))
print(np.pi)
print(np.log(1))
print(np.exp(0))
```

```
1.0
0.0
3.141592653589793
0.0
1.0
```

And so much more! RTFM!

Numpy has **many more functions and tools** to offer!

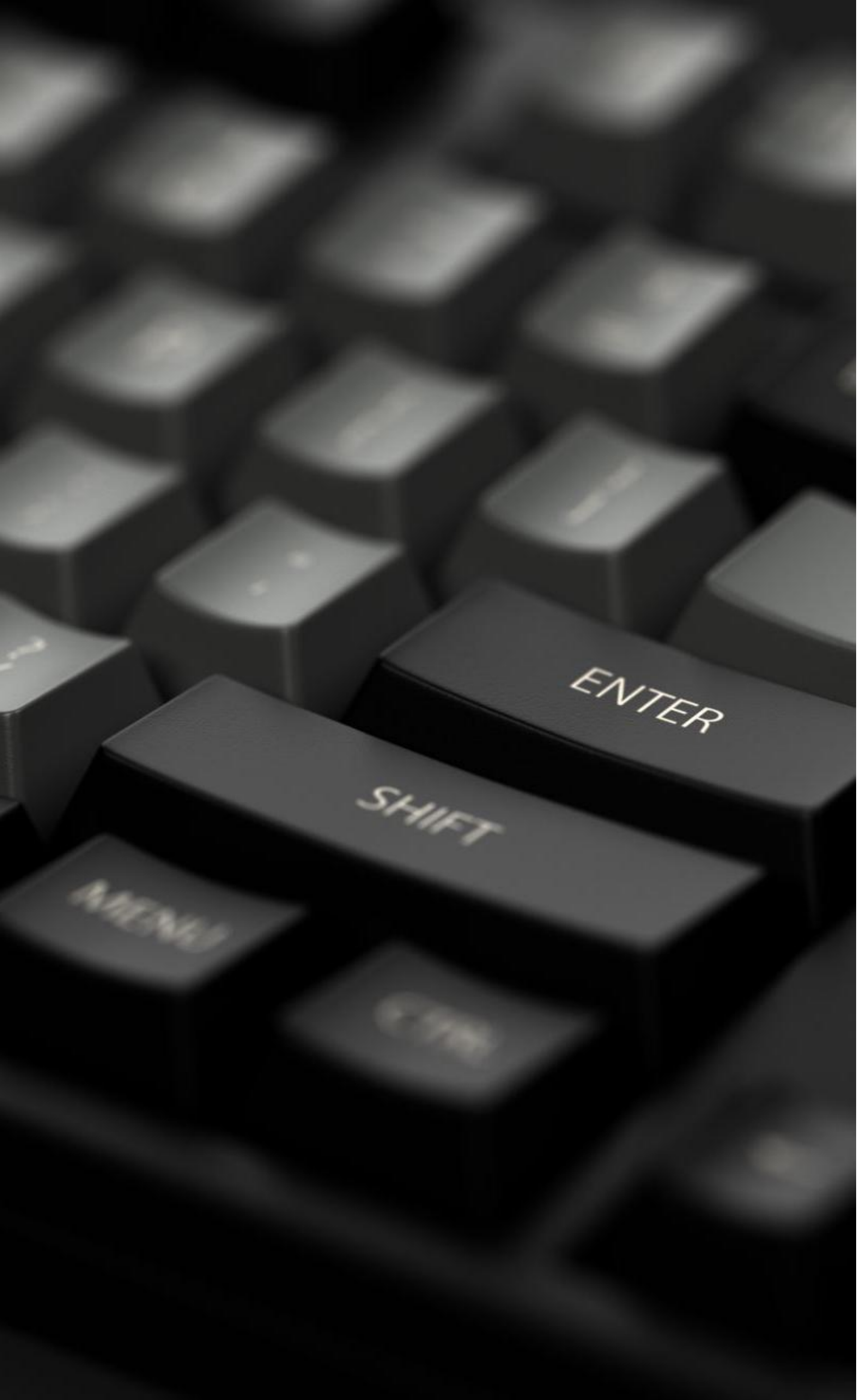
- E.g., **Random functions** (to be covered in an upcoming session, if time allows?)

```
# Numpy.random.choice() mimics a dice roll
dice_roll = np.random.choice([1, 2, 3, 4, 5, 6])
print(dice_roll)
dice_roll = np.random.choice([1, 2, 3, 4, 5, 6])
print(dice_roll)
dice_roll = np.random.choice([1, 2, 3, 4, 5, 6])
print(dice_roll)
dice_roll = np.random.choice([1, 2, 3, 4, 5, 6])
print(dice_roll)
dice_roll = np.random.choice([1, 2, 3, 4, 5, 6])
print(dice_roll)
dice_roll = np.random.choice([1, 2, 3, 4, 5, 6])
print(dice_roll)
```

```
4
6
1
4
3
2
```

- Learn more about Numpy (RTFM!) here:

https://numpy.org/doc/stable/



Conclusion (Chapter 12)

- A quick word on memory management and lists: **aliasing**, **shallow** and **deep copies.**
- What is a **library**?
- What is the **Numpy library** and its basic concepts (e.g. arrays, math functions, etc.)?

Let us move on to our first Mini-project now!

Wait for me to explain and show you first.